

CONTACT

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PROFESSIONAL EXPERIENCE

AI Engineer | August 2025 – Present

TIA S.A

- Development of a chatbot using RAG techniques for more precise context management between documents.
- Creation of algorithms with OCR, VLM, and LLM models to parse documents more accurately before creating a RAG.
- Development of intelligent agents for the sales and customer service departments, which access tables to extract and analyze information.

Robotics and AI Engineer | February 2025 – November 2025

CEDIA

- Development of a real-time voice chat between a humanoid robot and a person using local STT, LLM, and TTS models.
- Development of a RAG using LLM techniques to obtain relevant information about the person with whom the robot will converse.
- Development of Computer Vision algorithms to detect emotions, facial and body points, and to run various games with the robot.

Digital Factory Intern | August 2024 – January 2025

Unilever

- Development of machine learning algorithms for predicting product manufacturing processes.
- Planning robotics and computer vision projects.
- Design and implementation of applications to automate internal records.

Software Developer for UGV | February 2024 – May 2024

SUNY (South Korea – Incheon)

- Development of a follower robot using only a 2D LIDAR sensor, with speed control for precise object tracking.
- Migration of a robot from ROS1 to ROS2, ensuring a smooth and efficient transition.
- Consulting on robotics issues, providing technical support and recommendations on robotic systems.

Robotics Software Developer | April 2023 – September 2024

RAMEL

- Development of a robot control system using LLM models to configure tasks and routines using natural language.
- Development of a multi-robot algorithm based on master-slave methodologies using ROS2 Humble.
- Design, coordination, and development of robotics projects using ROS (ROS1 and ROS2).

LANGUAGES

English – Advanced Spanish – Native

SKILLS**LLM**

LangChain
LangGraph / LangSmith
LlamaIndex
Ollama / LlamaCpp

Computer Vision

PyTorch
TensorFlow
OpenCV
MediaPipe

Programming Languages

Python
SQL
JavaScript
C++

EDUCATION**Escuela Superior Politecnica del Litoral | 2019–2024**

Mechatronics Engineering

PROJECTS - LLM**Human-robot voice chat**

Creating a chat that enables fluid conversations between a bot and a human, the system uses Vosk STT, LlamaCpp with an optimized Llama3.2 model, and Piper TTS. The conversation takes place in real time, and all models are local.

RAG for personalized conversations

Creating a RAG that allows for more natural conversations with the person, providing insight into various important details for more interesting conversation topics.

LLM model for performing robot tasks

Using local LLM models to provide task and routine instructions to robots, the system itself feeds a RAG database that provides information on the robots' different actions and states.

INVESTIGATION**Software Architecture and Simulation Interface for Autonomous Underwater Vehicles**

Paper presented at IEEE

Development of a Multi-Platform AI-Based Software Interface for the Accompaniment of Children

MDPI Multimodal Technologies and Interaction Journal

**OFFICIAL
AWARDS****Technical Reviewer of Track Mechatronic Systems**

Technical Reviewer – COLIM 2026 – Track Mechatronic Systems

Article Reviewer in the track of Robotics and Automation

Article Reviewer – IEEE ETCM 2025 – Track Robotics & Automation

Software Architecture and Simulation Interface for Autonomous Underwater Vehicles

Best paper – IEEE ETCM 2024 – Track Robotics & Automation